Frage 1
Richtig
Erreichte Punkte 1,50 von 1,50
Frage markieren

The LSTM (without bias units and forget gate) is defined as

$$z(t) = \tanh(Ux(t) + Ph(t-1))$$

$$i(t) = \sigma(Vx(t) + Qh(t-1))$$

$$c(t) = c(t-1) + z(t) \odot i(t)$$

$$o(t) = \sigma(Wx(t) + Rh(t-1))$$

$$h(t) = \tanh(c(t)) \odot o(t)$$

with input vectors $x(t)$, hidden activation vectors $h(t)$, memory cell state vectors $c(t)$, gate activation vectors $z(t), i(t), o(t)$, weight matrices P, Q, R, U, V, W. Let $L(t) = L(y(t), \hat{y}(t))$ denote the loss at time t and let $L = \sum_{t=1}^{T} L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\frac{\partial L}{\partial c(t)}$ is a column vector. The diag operator turns a vector into a diagonal matrix, i.e., $\mathrm{diag}((1,1)^\top) = I \in \mathbb{R}^{2\times 2}$ and $\odot$ denotes Hadamard's product. Which of the following statements are true?

- [x] a. Changing the definition of the cell state to $c(t) = c(t-1) \odot f(t) + z(t) \odot i(t)$, where $f(t)$ is another sigmoid-activated gate reintroduces the vanishing gradient problem to a certain extent. ✓
- [ ] b. The gradient of the loss with respect to the hiddens was originally proposed to be approximated by $\frac{\partial L}{\partial h(t)} = \frac{\partial L(t+1)}{\partial h(t)}$.
- [x] c. The gradient of the loss with respect to the memory cell is $\frac{\partial L}{\partial c(t)} = \frac{\partial L}{\partial c(t+1)} + o(t) \odot \tanh'(c(t)) \odot \frac{\partial L}{\partial h(t)}$. ✓
- [ ] d. The fact that $\frac{\partial h(t)}{\partial c(t)} = \mathrm{diag}(o(t) \odot \tanh'(c(t)))$ is a diagonal matrix is the key to how LSTM solves the vanishing gradient problem.

Your answer is correct.

Die richtigen Antworten sind: Changing the definition of the cell state to $c(t) = c(t-1) \odot f(t) + z(t) \odot i(t)$, where $f(t)$ is another sigmoid-activated gate reintroduces the vanishing gradient problem to a certain extent., The gradient of the loss with respect to the memory cell is $\frac{\partial L}{\partial c(t)} = \frac{\partial L}{\partial c(t+1)} + o(t) \odot \tanh'(c(t)) \odot \frac{\partial L}{\partial h(t)}$.

Frage 2
Richtig
Erreichte Punkte 1,00 von 1,00
Frage markieren

Which of the following statements about RNNs are true?

- [ ] a. A disadvantage of RNNs is that they can only be applied to sequences of a fixed length.
- [x] b. The number of RNN parameters is independent of the sequence length. ✓
- [ ] c. Many RNN architectures are Turing complete. This means that they can be trained to exactly imitate the behavior of any computer program.
- [x] d. An RNN is equivalent to a feedforward network applied to the current sequence element and an internal state representation that depends on earlier sequence elements. ✓
- [ ] e. One advantage of RNNs is that their computations can be easily parallelized over time steps.

Your answer is correct.

Die richtigen Antworten sind: An RNN is equivalent to a feedforward network applied to the current sequence element and an internal state representation that depends on earlier sequence elements., The number of RNN parameters is independent of the sequence length.

Frage 3
Richtig
Erreichte Punkte 1,00 von 1,00
Frage markieren

Consider the fully recurrent network architecture (without output activation and bias units) defined as

$$s(t) = Wx(t) + Ra(t-1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation vectors $a(t)$, activation function $f(\cdot)$ and parameter matrices R, W, V. Let θ denote the vector of all network parameters shared in time and let $\theta(t)$ denote their usage at time t. Further, let $L(t) = L(y(t), \hat{y}(t))$ denote the loss function at time t and let $L = \sum_{t=1}^{T} L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\frac{\partial L}{\partial \theta}$ is a column vector.

Which statements about RTRL are true?

- [ ] a. Schmidhuber's approach divides the input sequence into chunks of size N (number hidden units) and performs RTRL within these chunks. Then it uses BPTT to consolidate the gradients for these chunks.
- [x] b. RTRL computes the gradients $\frac{\partial L(t)}{\partial \theta}$ during the forward pass already. ✓
- [x] c. BPTT considers the recursion $\frac{\partial L}{\partial s(t)} = \frac{\partial s(t+1)}{\partial s(t)} \frac{\partial L}{\partial s(t+1)} + \frac{\partial L(t)}{\partial s(t)}$ while RTRL considers the recursion $\frac{\partial s(t)}{\partial \theta} = \frac{\partial s(t)}{\partial \theta(t)} + \frac{\partial s(t-1)}{\partial \theta} \frac{\partial s(t)}{\partial s(t-1)}$. ✓
- [ ] d. The term $\frac{\partial s(t)}{\partial \theta}$ generally has $O(N^4)$ elements, where N denotes the number of hidden units

Your answer is correct.

Die richtigen Antworten sind: BPTT considers the recursion $\frac{\partial L}{\partial s(t)} = \frac{\partial s(t+1)}{\partial s(t)} \frac{\partial L}{\partial s(t+1)} + \frac{\partial L(t)}{\partial s(t)}$ while RTRL considers the recursion $\frac{\partial s(t)}{\partial \theta} = \frac{\partial s(t)}{\partial \theta(t)} + \frac{\partial s(t-1)}{\partial \theta} \frac{\partial s(t)}{\partial s(t-1)}$., RTRL computes the gradients $\frac{\partial L(t)}{\partial \theta}$ during the forward pass already.

Frage 4
Falsch
Erreichte Punkte 0,00 von 1,00
Frage markieren

Which statements about LSTM dropout are correct?

- [ ] a. LSTM needs special dropout techniques because a naive application will reset any memory cell with exponentially high probability.
- [x] b. LSTM needs special dropout techniques because a naive application of dropout will result in vanishing/exploding gradients. ✗
- [x] c. Dropout is a regularization technique that randomly sets activations to zero during the forward pass. ✓
- [ ] d. The central idea of zoneout is to randomly clear the updates to the memory cell instead of its content.
- [ ] e. It is especially important to apply dropout when evaluating the model.

Your answer is incorrect.

Die richtigen Antworten sind: LSTM needs special dropout techniques because a naive application will reset any memory cell with exponentially high probability., The central idea of zoneout is to randomly clear the updates to the memory cell instead of its content., Dropout is a regularization technique that randomly sets activations to zero during the forward pass.

Frage 5
Teilweise richtig
Erreichte Punkte 0,33 von 2,00
Frage markieren

Consider the fully recurrent network architecture (without output activation and bias units) defined as

$$s(t) = Wx(t) + Ra(t-1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation vectors $a(t)$, activation function $f(\cdot)$ and parameter matrices R, W, V. Let $L(t) = L(y(t), \hat{y}(t))$ denote the loss function at time t and let $L = \sum_{t=1}^{T} L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\delta(t) = \frac{\partial L}{\partial s(t)}$ is a column vector. Which of the following statements are true?

- [x] a. BPTT is the standard algorithm for computing the gradients of a recurrent neural network. ✓
- [ ] b. BPTT has to store all hidden activations $a(1), a(2), \ldots, a(T)$, i.e., its memory demand grows with the sequence length.
- [x] c. The gradient of the loss with respect to the recurrent weights R can be written as $\frac{\partial L}{\partial R} = \sum_{t=1}^{T} \delta(t) s^{\top}(t-1)$. ✗
- [ ] d. The gradient of the loss with respect to the input weights W can be written as $\frac{\partial L}{\partial W} = \sum_{t=1}^{T} x(t) \delta^{\top}(t)$.
- [ ] e. The deltas fulfill the recursive relation $\delta(t) = \operatorname{diag}(f'(s(t))) \left(V \frac{\partial L(t)}{\partial \hat{y}(t)} + R\delta(t+1) \right)$.

Your answer is partially correct.

Sie haben 1 richtig ausgewählt.
Die richtigen Antworten sind: BPTT is the standard algorithm for computing the gradients of a recurrent neural network., BPTT has to store all hidden activations $a(1), a(2), \ldots, a(T)$, i.e., its memory demand grows with the sequence length.

Frage 6
Richtig
Erreichte Punkte 1,00 von 1,00
Frage markieren

Which statements about LSTM applications are true?

- a. LSTM can be used for protein classification. Hereby, the protein is represented as a sequence of atoms and the labels are either functional or structural properties of the respective protein.
- b. "word2vec" (Mikolov et al., 2013) is an example for a word embedding technique. ✓
- c. Fan et al. (2015) trained an LSTM to predict audio from videos of speeches. Given a muted video recording of a speech, they could accurately recover the sound of the speech.
- d. A word embedding space is a real vector space in which words are represented as vectors (points) which often show meaningful relations to each other. For instance, let $f(\cdot)$ be the (trained) embedding function, then $f(\text{queen}) - f(\text{woman}) + f(\text{man})$ is found close to the point $f(\text{king})$. ✓
- e. Radford et al. (2017) used an LSTM to discover sentiment in Amazon reviews by training it for sequence classification (good vs. bad) on the review text.

Your answer is correct.

Die richtigen Antworten sind: A word embedding space is a real vector space in which words are represented as vectors (points) which often show meaningful relations to each other. For instance, let $f(\cdot)$ be the (trained) embedding function, then $f(\text{queen}) - f(\text{woman}) + f(\text{man})$ is found close to the point $f(\text{king})$., "word2vec" (Mikolov et al., 2013) is an example for a word embedding technique.

Frage 7
Teilweise richtig
Erreichte Punkte 0,17 von 1,00
Frage markieren

Which statements about the self-attention mechanism are correct?

- a. The vector at position t in the output sequence of a self-attention layer is computed by summing the value vectors, each weighted by its corresponding attention score from the t-th row of the attention matrix.
- b. The Multi-head Attention mechanism consists of a total of four weight matrices. ✓
- c. In the standard self-attention formulation, cosine similarity is used as a measure of alignment between keys and values.
- d. Similar to the hidden state in an LSTM, the attention matrix is updated in a recurrent manner for each token in an input sequence.
- e. The raw attention scores are normalized by applying a sigmoid activation function row-wise to the attention matrix. ✗

Your answer is partially correct.

Sie haben 1 richtig ausgewählt.
Die richtigen Antworten sind: The vector at position t in the output sequence of a self-attention layer is computed by summing the value vectors, each weighted by its corresponding attention score from the t-th row of the attention matrix., The Multi-head Attention mechanism consists of a total of four weight matrices.

Frage 8
Teilweise richtig
Erreichte Punkte 1,00 von 1,50
Frage markieren

Backpropagation through time (BPTT) is the standard gradient computation algorithm for RNNs. However, there exist alternatives, e.g., truncated BPTT and real-time recurrent learning (RTRL). Which statements about these algorithms are true.

- a. Truncated BPTT has advantages if the input sequences are short.
- b. RTRL computes the same gradients as BPTT ✓
- c. Truncated BPTT computes the same gradients as BPTT.
- d. RTRL is very slow and therefore rarely used in practice. ✓
- e. The runtime complexity of RTRL is $O(N^3)$ per time step, where N denotes the number of hidden units.
- f. Truncated BPTT divides the input sequence into chunks and propagates the deltas only within chunks but not across them.

Your answer is partially correct.

Sie haben 2 richtig ausgewählt.
Die richtigen Antworten sind: Truncated BPTT divides the input sequence into chunks and propagates the deltas only within chunks but not across them., RTRL computes the same gradients as BPTT, RTRL is very slow and therefore rarely used in practice.

Frage **9**

Teilweise richtig

Erreichte Punkte 1,20 von 1,50

Frage markieren

Which statements about the xLSTM are true?

- [x] a. The mLSTM matrix memory size is independent of sequence length. ✓
- [ ] b. The memory capacity of xLSTM increases with the sequence length.
- [x] c. The mLSTM variant of xLSTM uses a matrix memory and covariance update rule. ✓
- [ ] d. The sLSTM cell doesn't have recurrent connections.
- [x] e. The sLSTM memory is independent of sequence length and hidden dimension. ✓
- [ ] f. The mLSTM cell doesn't have recurrent connections.
- [x] g. Exponential gating addresses the limitation of LSTMs regarding their inability to revise storage decisions - it allows xLSTM to scale down older inputs (revise past storage decisions) ✓

Die Antwort ist teilweise richtig.

Sie haben 4 richtig ausgewählt.

Die richtigen Antworten sind: Exponential gating addresses the limitation of LSTMs regarding their inability to revise storage decisions - it allows xLSTM to scale down older inputs (revise past storage decisions), The mLSTM variant of xLSTM uses a matrix memory and covariance update rule., The sLSTM memory is independent of sequence length and hidden dimension., The mLSTM matrix memory size is independent of sequence length., The mLSTM cell doesn't have recurrent connections.

Frage **10**

Richtig

Erreichte Punkte 1,00 von 1,00

Frage markieren

The Elman Network (without output activation function and bias units) can be defined as

$s(t) = Wx(t) + a(t-1)$

$a(t) = f(s(t))$

$\hat{y}(t) = Va(t)$

with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation vectors $a(t)$, activation function $f(\cdot)$, and parameter matrices W, V. Which of the following statements are true?

- [ ] a. The matrix W contains the recurrent weights.
- [x] b. The recurrent weights are equal to the identity matrix. ✓
- [x] c. The recurrent Jacobian of the hidden activations is $\frac{\partial a(t)}{\partial a(t-1)} = \mathrm{diag}(f'(s(t)))$. ✓
- [ ] d. The hidden units of the Elman network are interconnected, i.e., at time t each hidden unit receives input from all other hidden units at time $t-1$.
- [x] e. The Elman network is well suited for training with RTRL because the recurrent Jacobian has a diagonal structure and this reduces the computational complexity. ✓

Your answer is correct.

Die richtigen Antworten sind: The recurrent weights are equal to the identity matrix., The recurrent Jacobian of the hidden activations is $\frac{\partial a(t)}{\partial a(t-1)} = \mathrm{diag}(f'(s(t)))$., The Elman network is well suited for training with RTRL because the recurrent Jacobian has a diagonal structure and this reduces the computational complexity.

Frage 11

Teilweise richtig

Erreichte Punkte 0,50 von 1,00

Frage markieren

Which statements about LSTM are true?

- a. The forget gate should be initialized with negative bias values.
- b. The forget gate solves the vanishing gradient problem.
- c. Most LSTM networks used in practice feature a forget gate. ✓
- d. The focused LSTM network feeds the input vector $x(t)$ only into the block input $z(t)$ and not into input gate $i(t)$ or output gate $o(t)$.
- e. The inputs to the lightweight LSTM have to fulfill the Markov property.
- f. An LSTM with coupled input gate and forget gate, i.e., $c(t) = (1 - i(t)) \odot c(t-1) + i(t) \odot z(t)$, leads (alongside other changes) to the gated recurrent unit (GRU).
- g. "Ticker steps" refers to the technique of letting an LSTM run for more time steps even after the input sequence has been taken in by the network. ✓
- h. To counter the drift effect, the input gate should be initialized with positive values.

Your answer is partially correct.

Sie haben 2 richtig ausgewählt.
Die richtigen Antworten sind: Most LSTM networks used in practice feature a forget gate., An LSTM with coupled input gate and forget gate, i.e., $c(t) = (1 - i(t)) \odot c(t-1) + i(t) \odot z(t)$, leads (alongside other changes) to the gated recurrent unit (GRU)., The focused LSTM network feeds the input vector $x(t)$ only into the block input $z(t)$ and not into input gate $i(t)$ or output gate $o(t)$., "Ticker steps" refers to the technique of letting an LSTM run for more time steps even after the input sequence has been taken in by the network.

Frage 12

Teilweise richtig

Erreichte Punkte 0,67 von 1,00

Frage markieren

Which of the following statements are true regarding the Mamba architecture?

- a. Mamba's memory requirements scale quadratically $\mathcal{O}(n^2)$ with input sequence length.
- b. The Mamba architecture combines the H3 block of SSM with the MLP block structure of transformer. ✓
- c. Mamba uses pre-activation forget (sd), input (sb), and output (sc) gates - these gates control the flow of information within the Mamba block. ✓
- d. Mamba is considered a selective State Space Model (SSM)
- e. Mamba's selective SSM differs from traditional SSMs because its ($\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$) functions are independent of the input.

Die Antwort ist teilweise richtig.

Sie haben 2 richtig ausgewählt.
Die richtigen Antworten sind: Mamba is considered a selective State Space Model (SSM), The Mamba architecture combines the H3 block of SSM with the MLP block structure of transformer., Mamba uses pre-activation forget (sd), input (sb), and output (sc) gates - these gates control the flow of information within the Mamba block.

Frage 13

Falsch

Erreichte Punkte 0,00 von 1,00

Frage markieren

Which statements about attention mechanism are true?

- a. An LSTM with self-attention has access to all previous hidden states and cell states via a softmax-based attention mechanism.
- b. A spatial attention mechanism causes the network to treat each pixel in an image as equally important.
- c. The LSTM gates can be interpreted as a temporal attention mechanism. ✓
- d. The attention score $e(i)$ is obtained from the attention vector $a(i)$ via the softmax function. ✗
- e. The scaled dot-product attention mechanism generally has the form $\mathrm{softmax}\left(\frac{1}{\sqrt{d}}QK^\top\right)V$, where $K, Q, V \in \mathbb{R}^{L \times d}$ are keys, queries, and values, respectively, and L is the sequence length and d is the hidden dimension.

Your answer is incorrect.

Die richtigen Antworten sind: The LSTM gates can be interpreted as a temporal attention mechanism., An LSTM with self-attention has access to all previous hidden states and cell states via a softmax-based attention mechanism., The scaled dot-product attention mechanism generally has the form $\mathrm{softmax}\left(\frac{1}{\sqrt{d}}QK^\top\right)V$, where $K, Q, V \in \mathbb{R}^{L \times d}$ are keys, queries, and values, respectively, and L is the sequence length and d is the hidden dimension.

?

Frage 14
Teilweise richtig
Erreichte Punkte 0,67 von 1,00
Frage markieren

Which of the following statements about LSTM variants are true?
(Note: Citations are provided for clarification only and do not need to be verified.)

- [x] a. A Bidirectional LSTM (Schuster et al., 1997) has advantages over the unidirectional version if the input sequences do not have a preferred processing direction. ✓
- [ ] b. Peephole connections (Gers and Schmidhuber, 2000) provide gates with direct read access to the cell state.
- [ ] c. The key idea of multidimensional LSTM (Graves et al., 2007) is to enable the network to process sequences consisting of vectors which stem from high-dimensional vector spaces.
- [x] d. Convolutional LSTM (Shi et al., 2015) can process sequences of images, e.g., videos. ✓
- [ ] e. Stacked LSTM (Graves et al., 2013) stack LSTM layer by using one LSTM's final cell state as the initial cell state for the next LSTM.

Your answer is partially correct.

Sie haben 2 richtig ausgewählt.
Die richtigen Antworten sind: Peephole connections (Gers and Schmidhuber, 2000) provide gates with direct read access to the cell state., A Bidirectional LSTM (Schuster et al., 1997) has advantages over the unidirectional version if the input sequences do not have a preferred processing direction., Convolutional LSTM (Shi et al., 2015) can process sequences of images, e.g., videos.

Frage 15
Teilweise richtig
Erreichte Punkte 0,17 von 1,00
Frage markieren

Which statements about the gated recurrent unit (GRU, Cho et al., 2014) are true?

- [ ] a. In the GRU, a so-called "reset gate" $r(t)$ regulates how strongly the increments $z(t)$ depend on the old hiddens $h(t-1)$.
- [x] b. The GRU has one gate less than the fully featured LSTM architecture. ✓
- [ ] c. The GRU does not have any form of forget-gate mechanism.
- [x] d. The so-called "update gate" of the GRU acts as a coupled input and output gate. ✗
- [x] e. The standard GRU has less parameters than a fully featured LSTM (i.e., with forget gate) that has the same number of hidden units. ✓

Your answer is partially correct.

Sie haben 2 richtig ausgewählt.
Die richtigen Antworten sind: The GRU has one gate less than the fully featured LSTM architecture., The standard GRU has less parameters than a fully featured LSTM (i.e., with forget gate) that has the same number of hidden units., In the GRU, a so-called "reset gate" $r(t)$ regulates how strongly the increments $z(t)$ depend on the old hiddens $h(t-1)$.

Frage **16**

Teilweise richtig

Erreichte Punkte 0,25 von 1,50

Frage markieren

The LSTM (without bias units and forget gate) is defined as

$$z(t) = \tanh(Ux(t) + Ph(t-1))$$

$$i(t) = \sigma(Vx(t) + Qh(t-1))$$

$$c(t) = c(t-1) + z(t) \odot i(t)$$

$$o(t) = \sigma(Wx(t) + Rh(t-1))$$

$$h(t) = \tanh(c(t)) \odot o(t)$$

with input vectors $x(t)$, hidden activation vectors $h(t)$, memory cell state vectors $c(t)$, gate activation vectors $z(t), i(t), o(t)$, weight matrices P, Q, R, U, V, W. Let $L(t) = L(y(t), \hat{y}(t))$ denote the loss at time t and let $L = \sum_{t=1}^{T} L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\frac{\partial L}{\partial c(t)}$ is a column vector. The diag operator turns a vector into a diagonal matrix, i.e., $\mathrm{diag}((1,1)^\top) = I \in \mathbb{R}^{2\times 2}$ and $\odot$ denotes Hadamard's product. Which of the following statements are true?

- a. Because of the simple structure of the memory cell, the LSTM architecture fails to be Turing complete.
- b. The memory cell fulfills $\frac{\partial c(t)}{\partial c(t-\tau)} = I$ (neglecting dependencies via the hiddens) for any $\tau \in \{1, \dots t-1\}$, where I is the identity matrix. This solves the vanishing gradient problem and is called constant error carousel. ✓
- c. The LSTM architecture has no exploding or vanishing gradients because $\frac{\partial h(t)}{\partial h(t-1)}$ is a diagonal matrix. ✗
- d. If we adapt $z(t) = \sigma(Ux(t) + Ph(t-1))$, then $z(t) \geq 0$ for all t and the memory cells will always increase at every time step. This can be a problem for very long sequences but can also be helpful for certain problems.
- e. The gradient of the loss with respect to the hiddens is $\frac{\partial L}{\partial h(t)} = \frac{\partial L(t)}{\partial h(t)} + U\frac{\partial L}{\partial z(t-1)} + V\frac{\partial L}{\partial i(t-1)} + R\frac{\partial L}{\partial o(t-1)}$.

Your answer is partially correct.

Sie haben 1 richtig ausgewählt.

Die richtigen Antworten sind: The memory cell fulfills $\frac{\partial c(t)}{\partial c(t-\tau)} = I$ (neglecting dependencies via the hiddens) for any $\tau \in \{1, \dots t-1\}$, where I is the identity matrix. This solves the vanishing gradient problem and is called constant error carousel., If we adapt $z(t) = \sigma(Ux(t) + Ph(t-1))$, then $z(t) \geq 0$ for all t and the memory cells will always increase at every time step. This can be a problem for very long sequences but can also be helpful for certain problems.

Frage **17**

Teilweise richtig

Erreichte Punkte 0,17 von 1,00

Frage markieren

Which statements about the self-attention mechanism are correct?

- a. The Multi-head Attention mechanism consists of a total of three weight matrices.
- b. The vector at position t in the output sequence of a self-attention layer is computed by summing the key vectors, each weighted by its corresponding attention score from the t-th row of the attention matrix.
- c. In contrast to recurrent neural networks, self-attention processes input sequences in parallel.
- d. In the standard self-attention formulation, the euclidean distance is used as a measure of alignment between keys and queries. ✗
- e. The raw attention scores are normalized by applying a softmax activation function row-wise to the attention matrix. ✓

Your answer is partially correct.

Sie haben 1 richtig ausgewählt.

Die richtigen Antworten sind:
In contrast to recurrent neural networks, self-attention processes input sequences in parallel.,
The raw attention scores are normalized by applying a softmax activation function row-wise to the attention matrix.

Überprüfung beenden